Pandas Dataframe

Assignment

PW SKILLS

Instructions for Students

1. Create a new Google Colab notebook and complete the assignment using Python and Pandas.
2. For each question:
   - Write the required Python code in a code cell.
   - Run the cell to generate the output.
3. Make sure your notebook contains:
   - The question statement
   - The Python code used to solve the question
   - The output generated after running the code
4. Do not delete outputs. Your final notebook should show both code and corresponding outputs.
5. Add Markdown cells to clearly label each question.

Q1. Creating a DataFrame

Create a Pandas DataFrame using a Python dictionary with the following columns:

- Product
- Region
- Sales
- Quantity

Add at least 5 rows of sample data and display the DataFrame.

Dataset for Questions 2–10

Use the dataset file provided:

Retail_Dataset

Load the dataset:

```
import pandas as pd

df = pd.read_csv("Retail_Dataset.csv")
```

Q2. Data Inspection

Perform the following inspection tasks:

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows.
3. Show the structure of the dataset using info().
4. Display the shape of the dataset.

Q3. Column Selection

Select and display only the following columns:

- Product
- Region
- Sales

Q4. Indexing using .loc

Using .loc, perform the following:

1. Display the row with index 10.
2. Display the Sales value of the row with index 20

Q5. Indexing using .iloc

Using .iloc, display:

1. The first 6 rows and first 4 columns.
2. The Sales value from the 3rd row.

Q6. Row Slicing

Display:

1. The first 10 rows of the dataset.
2. Rows 15 to 25.

Q7. Filtering Data

Filter the dataset to display:

1. Rows where Region = "East".
2. Rows where Sales > 1500.

Q8. Multiple Condition Filtering

Display rows where:

- Region is West
- Sales is greater than 1200

Q9. Descriptive Statistics

Generate summary statistics using:

describe()

From the result identify:

- Mean sales value
- Maximum quantity sold
- Minimum sales value

Q10. Aggregation Analysis

Using the Sales column, calculate:

1. Total sales
2. Average sales
3. Maximum sales
4. Minimum sales